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Research paper

Essential system response information for transient-based leak detection methods

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ABSTRACT

Many transient-based techniques have been developed for leak detection in piping fluid systems, and they are thought to use leak-induced damping and reflection effects to detect and locate the problem. This work investigates the relative importance of the damping and reflective effects from the leak to the leak detection methods with the importance of both effects tested using four representative methods, which vary significantly in their modes of operation and provide the coverage of the major developments in the transient leak detection field. It was found that all transient-based methods are unable to detect and locate the leak in the absence of the leak-reflected signal, whereas the leak is detected and located even in the absence of leak-induced damping. The results indicate that the information previously thought to be contained in the damping of slow transients was found to be low frequency reflections from the leak.

Keywords: Inverse transient analysis, leak detection, pipe system, system response function, transient damping, transient reflection

1 Introduction

Compared with many commercial leak detection techniques, the application of transient-based methods has advantages in terms of their speed, ability to work online and large operational range (Wang 2002). A leak in a pipeline system will result in an increased transient damping rate and the creation of new leak-reflected signals within the time trace. Many transient-based leak detection methods have been developed and applied to water piping systems using information contained within these two effects. The developed leak detection methods vary greatly in their modes of operation, but may be divided into four main categories, namely the transient reflection method (TRM), the transient damping method (TDM), the system

response function method (SRFM) and the inverse transient method (ITM). It was found through recent field studies that all transient-based leak detection methods face difficulties in systems with a high level of complexities. A better understanding of their dependency on the different types of information contained within the transient trace is essential when designing new transient signals for improving their accuracy.

In general, all current transient-based leak detection methods are unable to deal with systems with a high level of complexity (practical field level), and the understanding of their dependency on the different types of information contained within the transient trace is essential for improving their accuracy. A recent literature review of Colombo *et al.* (2009) indicates that the validations of the present transient-based leak detection

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methods are mainly focused on simple systems, and many had not yet involved field situations. One of the reasons is the potential difficulty of dealing with practical factors affecting the transient damping and reflections such as external fluxes, junction connections, elbow connections and other system properties. This study aims to investigate the relative importance of the transient information (damping and reflections) to the application of the transient-based leak detection methods and to inspect the impact of these practical factors on the accuracy of these methods. This research studies the accuracy of the four types of transient leak detection methods if the signal contains both leak-induced damping and reflection information compared to when only one of the two types of information exists. The latter scenarios are created through modifying the original transient signal, and this process is described below.

2 Investigation methods

2.1 One-dimensional waterhammer model

The classical one-dimensional (1D) waterhammer model is adopted herein, mathematically expressed as (Chaudhry 1987, Ghidaoui *et al.* 2005)

$$\frac{gA}{a^2} \frac{\partial H}{\partial t} + \frac{\partial Q}{\partial x} = 0 \quad (1)$$

$$\frac{\partial Q}{\partial t} + gA \frac{\partial H}{\partial x} + \frac{f}{2DA} Q|Q| = 0 \quad (2)$$

where H is the pressure head, Q the pipe discharge, A the pipe cross-sectional area, D the pipe diameter, a the wave speed, x the spatial coordinate along the pipeline, t the time, g the gravitational acceleration and f the pipe friction factor.

Only steady friction is considered in the application tests herein. Previous works have shown that unsteady friction creates additional pressure head damping in addition to steady friction and results in a more rapid decay of information content in the transient signal (Chaudhry 1987, Ghidaoui *et al.* 2005). As the present aim is to compare the relative importance of the two types of leak-induced information in a transient trace, it is advantageous to maximize the amount of information contained in the trace for analysis. It is important to note that the present findings are still valid in systems with unsteady friction. Nixon *et al.* (2006) and Nixon and Ghidaoui (2007) studied the impact of unsteady friction effect on the leak detection method and used the TDM as an example. Their results indicate that there is more amplitude damping of the frequency modes if the unsteady friction effect is included in the TDM and the unsteady friction effect has no impact on the accuracy of the technique as long as it is correctly represented in the numerical outputs.

2.2 Transient-based leak detection methods

Four types of transient-based leak detection methods are used herein to investigate the essential system response information for locating leaks in pipe systems. For clarity, four representations of these methods are briefly reviewed in this section. Among the developed methods, TRM is the easiest to apply and the leak is located through the reflection information of the pressure signal. Brunone (1999) introduced the method with the following equation

$$x_L^* = \frac{a}{2L} (t_2 - t_1) \quad (3)$$

where x_L^* is the dimensionless leak location representing the distance of the leak from the upstream boundary normalized by the pipe length, L the pipe length, t_1 the instant at which the pressure wave generated at the end valve arrives at the measurement location and t_2 the instant at which the reflected wave at the leak reaches the measurement location. Brunone and Ferrante (2001) conducted an experimental validation of the technique, and the TRM was able to accurately predict the leak location by measuring the arrival times of the signal reflections. Other works that improved on the application of this idea include the introduction of change detection algorithm such as wavelet (Ferrante *et al.* 2007, 2009) and cumulative sum (Lee *et al.* 2007). While TRM relies solely on the reflection information within the transient trace, an alternative method was developed, which was designed to work solely on the damping rate of the transient signal.

The TDM as proposed by Wang *et al.* (2002) is a novel leak detection method utilizing the relative damping rates of the first two harmonic frequency components in the pressure head trace to locate the problem. This method was derived analytically from the 1D waterhammer model for a single pipe system. The equation relating to the harmonic damping ratios and the leak location is

$$\frac{R_{n1L}}{R_{n2L}} = \frac{\sin^2(n_1 \pi x_L^*)}{\sin^2(n_2 \pi x_L^*)} \quad (4)$$

where R_{nL} is the leak-induced damping rate for the n th mode and n_i is the mode number, with $i = 1$ or 2 . The TDM assumes system linearity and only applies to single pipeline systems with weak unsteady friction. These assumptions were verified using a two-dimensional transient model by Nixon *et al.* (2006), indicating that the assumptions did not limit the model applicability in practice if the system was simple and the friction damping effect was represented correctly in pressure head traces.

The TRM and TDM were designed to use only one of the two types of leak information contained within a transient signal. There are other methods such as SRFM that uses the entire transient signal to detect and locate the leak. Liou and Tian (1995), Liou (1998), Ferrante and Brunone (2003), Covas *et al.* (2005), Lee *et al.* (2006, 2007) and Sattar and Chaudhry

(2008) located the leak in the system by analysing the harmonic and impulse modes of pressure wave traces, and the SRFM presented by Lee *et al.* (2006, 2007) was adopted therein. The main principle of the technique is shown in Fig. 1.

In any system, a transient signal can be considered as the result of an initial flow disturbance imposed on the system (input x), whereas the measured pressure response is the output from system y . The behaviour of a pipeline system can then be described as a function that produces the output for a given input. The relationship between input and output signals in the time domain is given as a convolutional integral (Lee *et al.* 2007)

$$y(t) = \int_{-\infty}^{+\infty} x(t^*)I(t-t^*)dt^* \quad (5)$$

where $I(\cdot)$ is the impulse response function (IRF) of the system containing all information pertaining to the behaviour of the system and t^* is the historical time point. By applying the Fourier transform (Kreyszig 1993), the expression becomes

$$Y(\omega) = X(\omega) \cdot F(\omega) \quad (6)$$

where $F(\cdot)$ is the frequency response function (FRF) of the system, ω the frequency, $X(\cdot)$ the Fourier transform of the input signal x and $Y(\cdot)$ the Fourier transform of the output signal y .

The system response function (either in the time domain as the IRF or in the frequency domain as the FRF) describes the fundamental response of the system from an impulse excitation. A leak in the pipeline results in a change in this system response, for example, the FRF from leaking and non-leaking pipelines (Fig. 2). In an intact pipeline, the FRF consists of a series of uniformly-spaced and -sized harmonic peaks (thin dashed line in

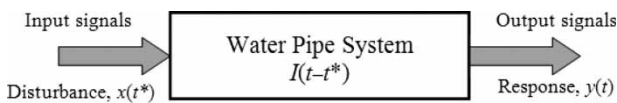


Figure 1 Schematic for SRFM

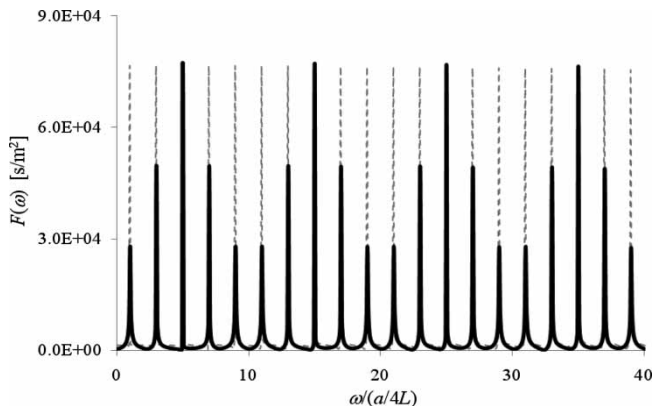


Figure 2 Converted FRF results in the frequency domain: —, intact pipe response; —, leaking pipe response

Fig. 2), but in a leaking system, the sizes of these peaks vary with frequency (bold solid line in Fig. 2) termed as “leak-induced pattern” by Lee *et al.* (2006, 2007). This perturbation pattern is the result of leak-induced changes to the transient response and can be used to locate the leak. Lee *et al.* (2006) found the analytical expression for the leak-induced pattern as

$$\hat{h} = \alpha \cos(2\pi m x_L^* - \theta) + \beta \quad (7)$$

where $\hat{h}$ is the inverted peak head magnitude from FRF result. α , θ , β are the fitting coefficients and m is the peak magnitude number. The coefficient x_L^* is a measure of the leak location from the upstream boundary. Lee *et al.* (2006) demonstrated that this method agrees well with experimental results. As the technique uses the entire transient signal, it has been suggested that this technique relies on both the reflection and damping information from the leak.

Another popular method that uses the entire transient signal is ITM, which operates by calibrating and matching the outputs from a numerical model to measured data records (Liggett and Chen 1994, Brunone 1999, Vítkovský *et al.* 2000, Al-Khomairi 2008, Shamloo and Haghighi 2009). The predicted response with the leaks in the correct locations results in the closest match with the measured results. Since ITM uses the entire transient trace for calibration, this method is thought to utilize both the leak-induced damping and reflection information. The objective function for optimization can be expressed as

$$\min : G = \sum_{i=1}^N |H_i^m - H_i^p| \quad (8)$$

where G is the fitness of the objective function, H^m the measured pressure head, H^p the predicted pressure head by using numerical models, $i = 1, \dots, N$ the time-step point for comparison and N the total number of data points. Various mathematical algorithms have been used for the objective optimization function of Eq. (8) (Liggett and Chen 1994, Vítkovský *et al.* 2000, Al-Khomairi 2008, Shamloo and Haghighi 2009). Herein, the genetic algorithm is adopted as proposed by Vítkovský *et al.* (2000), who applied ITM successfully in a hypothetical simple network for leak calibration, while their further study by laboratory experimental validation indicated that the success of this method depends on the system complexities such as boundary conditions and system properties (Vítkovský *et al.* 2007) being well-defined. Meanwhile, the computation load of ITM is the other difficulty for practical large-scale pipe systems.

3 Numerical experiments

The simple pipe system shown in Fig. 3 is used for the present numerical tests consisting of a constant head tank, a single pipeline and a boundary valve. The valve at the downstream boundary is initially fully closed and the transient is caused by a pulse

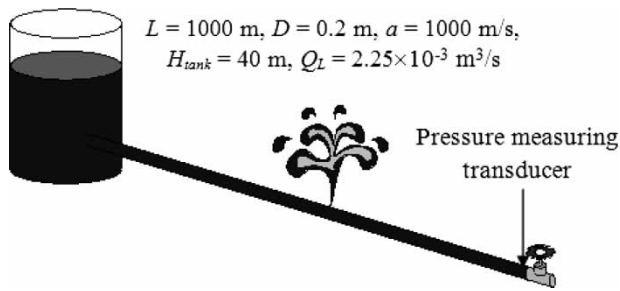


Figure 3 Schematic for pipe system with leakage

perturbation of the valve opening, resulting in the discharge variation Q_V shown in Fig. 4. A single leak was placed at various locations along the pipeline with the dimensionless leak location $x_L^* = 0.1, 0.3, 0.6, 0.8$ and 0.9 and initial leak discharge $Q_L = 2.25 \times 10^{-3} \text{ m}^3/\text{s}$. The pressure at the upstream valve face is collected for leak detection analysis. Note that the study identifies the relative importance of leak reflection and damping effects, and the analysis of different leak sizes should only be relevant to this study if the leak size affects these two effects in different ways. Leaks of all sizes in the transient pipeline result in reflections, as well as damping, and it is logical to use a leak creating reflection and damping information that are clearly observable for this investigation. This study intentionally uses a large leak to highlight its effect as a small leak may induce effects that are so small that it can be contaminated by machine error.

The numerical simulation is conducted by using the method of characteristics for the 1D waterhammer model with 100 discrete sections along the pipeline. The leak is modelled by the orifice equation (Lee *et al.* 2006, 2007). The friction factor used is $f = 0.015$. The transient pressure head traces for the situations with and without leakage are shown in Fig. 5, and the relative pressure head $\Delta H_t / \Delta H_0$ represents the immediate transient pressure head relative to the steady state ΔH_t normalized by the initialized transient pressure head ΔH_0 . In the leaking pipeline trace, the increased damping of the pressure peak and the additional reflected pulse signals from the leakage in the pipeline can be clearly observed. To investigate the relative importance of the two leak-induced changes for leak detection, initially both

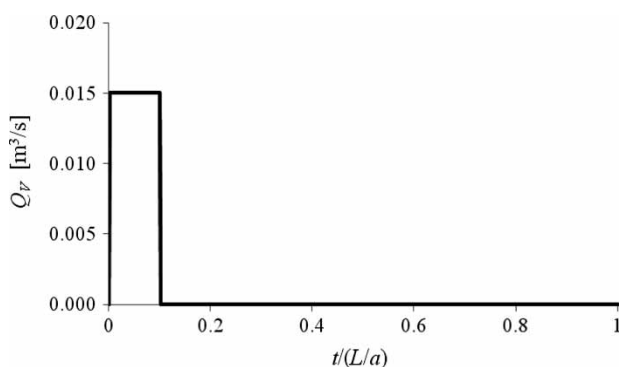


Figure 4 Valve discharge pulse with time

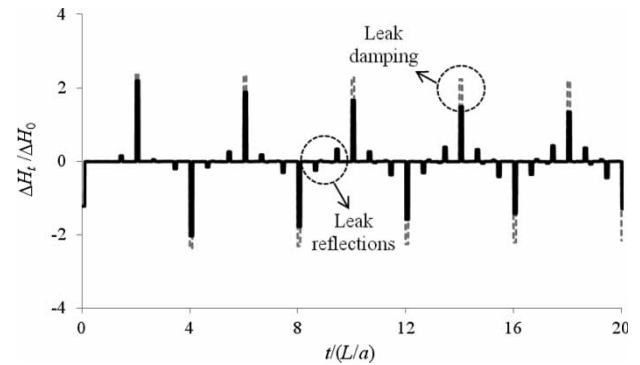


Figure 5 Transient head in the time domain for intact and leaking pipes: —, intact pipe response; —, leaking pipe response

effects are retained in the signal as different leak detection techniques are applied, followed by the situations in which only one of them will be considered in the absence of the other.

4 Results and discussion

4.1 Leak detection using leak damping and reflection

The leak detection results considering both damping and reflection information are listed in Table 1. The errors of prediction are calculated and listed in the last column of the result tables. For clarity, the calculation of prediction error is mathematically expressed as

$$\varepsilon(\%) = (x_{Lp}^* - x_{Lr}^*) \times 100 \quad (9)$$

where ε is the error of leak location prediction and x_{Lp}^* and x_{Lr}^* are the predicted and real dimensionless leak locations. The results indicate that the maximum relative errors of the leak location prediction for the four methods all are about 1%, which implies that all four methods can accurately predict the leak location using the original numerical transient traces from the model.

4.2 Leak detection using leak damping only and leak reflection only

To investigate the relative importance of leak-induced damping and reflection information for the leak detection techniques, the original transient signal was modified in two ways. In one case, the leak reflections are artificially removed from the data by replacing the reflected signal with the steady-state head, leaving only the leak-induced damping as shown in Fig. 6. Note that the modified signal in Fig. 6 damps at the same rate as the response from the leaking pipe. In the other case, the leak-induced damping of the signal can be removed by two alternatives: one is by adding the reflection information to the signal of the intact pipe such that the signal reflects at the same location as the response from a leaking pipe, and the other is

Table 1 Results if considering both leak damping and reflection information

Case	Real leak location, $x_L^*{}_r$	TRM		TDM		SRFM		ITM	
		$x_L^*{}_p$	ε (%)	$x_L^*{}_p$	ε (%)	$x_L^*{}_p$	ε (%)	$x_L^*{}_p$	ε (%)
1	0.10	0.10	0.0	0.10	0.0	0.10	0.0	0.10	0.0
2	0.30	0.30	0.0	0.29	-1.0	0.31	1.0	0.30	0.0
3	0.60	0.60	0.0	0.60	0.0	0.60	0.0	0.60	0.0
4	0.80	0.80	0.0	0.80	0.0	0.80	0.0	0.80	0.0
5	0.90	0.90	0.0	0.90	0.0	0.91	1.0	0.91	1.0

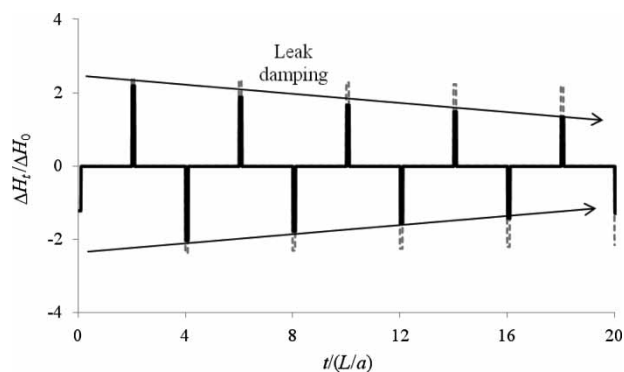


Figure 6 Transient head in the time domain for intact pipe and leaking pipe with leak damping only: ---, intact pipe response; —, leaking pipe response with leak reflections removed

by scaling up the amplitude of the main signals for the leaking pipe response such that the signal damps at the same rate as the response from an intact pipe, leaving only the leak reflection information. It was found that the two alternatives provide similar results. For illustration, Fig. 7 shows results pertaining to the first method and the obtained signal.

The leak detection results for the two situations are listed in Tables 2 and 3, and the errors of prediction for each case are also calculated. The numerical results in Table 2 indicate that if only leak damping information is present, significant errors result from all four leak detection methods with errors up to 40%. While this result is expected for the TRM, it is surprising

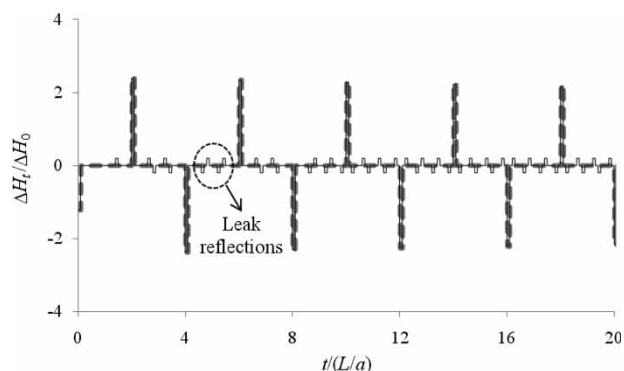


Figure 7 Transient head in the time domain for intact pipe and leaking pipe with leak reflection only: ---, intact pipe response; —, leaking pipe response with leak-damping removed

that TDM, which was thought to operate on the leak damping information, was also unable to locate the leak. In addition, techniques that use the entire transient signal for detection, such as SRFM and ITM, also failed to locate the leak if the leak reflections are removed.

In contrast, under the condition in which only leak reflection information is present, the predicted results by all four methods are significantly improved. Even for TDM previously thought to be a leak damping-based method, the maximum prediction error is 11% in this situation (as compared to 40% previously), indicating that TDM is more reliant on the presence of leak reflections rather than leak damping. This finding will be explained below. For the other three methods (TRM, SRFM and ITM), the leak location can be accurately obtained by using only the leak-induced reflections (Table 3) and the maximum errors for the leak location prediction are within 1%, which is similar to the accuracy obtained if both the leak damping and reflection information are present in the signal (Table 1). The absence of the leak damping information appears to have little impact on the operation or the accuracy of the leak detection methods.

The dependence of TDM on the leak reflection can be further illustrated by comparing the transient information utilized in TDM and the “leak-induced pattern” used in SRFM. The FRFs for the three situations (full leak-induced information, leak damping only and reflection only) are shown in Fig. 8. Figure 8(a) shows that the FRF for the case with the leak reflections removed has uniformly-spaced and -sized peaks for different harmonic frequencies (plotted as thin line), identical to FRF for the intact pipe case shown in Fig. 2. In contrast, the case with the leak-induced damping removed in Fig. 8(b) has peaks in exactly the same pattern as in the leaking response with full leak-induced information (despite differences in the absolute amplitudes for each peak due to a change in the overall energy of the signal). The comparative results demonstrate that leak-induced reflection alone can represent the leaking pattern in the FRF, and the leak damping plays no part in this information. This finding has further importance for TDM. The TDM works by taking the ratio between the first peak and the first trough in the above FRF results (refer to Eq. 4), and the removal of the leak-induced pattern in FRF removes all information required for TDM to operate. The TDM is in essence a leak reflection-based method, and the information previously thought to be

Table 2 Results by considering leak damping only

Case	Real leak location, $x_L^*{}_r$	TRM		TDM		SRFM		ITM	
		$x_L^*{}_p$	ε (%)	$x_L^*{}_p$	ε (%)	$x_L^*{}_p$	ε (%)	$x_L^*{}_p$	ε (%)
6	0.10	N/A	N/A	0.50	40.0	0.50	40.0	0.50	40.0
7	0.30	N/A	N/A	0.50	20.0	0.51	21.0	0.50	20.0
8	0.60	N/A	N/A	0.50	−10.0	0.50	−10.0	0.50	−10.0
9	0.80	N/A	N/A	0.50	−30.0	0.49	−31.0	0.50	−30.0
10	0.90	N/A	N/A	0.50	−40.0	0.50	−40.0	0.50	−40.0

N/A, not applicable.

Table 3 Results of considering leak reflection only

Case	Real leak location, $x_L^*{}_r$	TRM		TDM		SRFM		ITM	
		$x_L^*{}_p$	ε (%)	$x_L^*{}_p$	ε (%)	$x_L^*{}_p$	ε (%)	$x_L^*{}_p$	ε (%)
11	0.10	0.10	0.0	0.13	3.0	0.11	1.0	0.10	0.0
12	0.30	0.30	0.0	0.28	−2.0	0.31	1.0	0.30	0.0
13	0.60	0.60	0.0	0.71	11.0	0.60	0.0	0.60	0.0
14	0.80	0.80	0.0	0.85	5.0	0.79	−1.0	0.79	−1.0
15	0.90	0.90	0.0	0.84	−6.0	0.89	−1.0	0.90	0.0

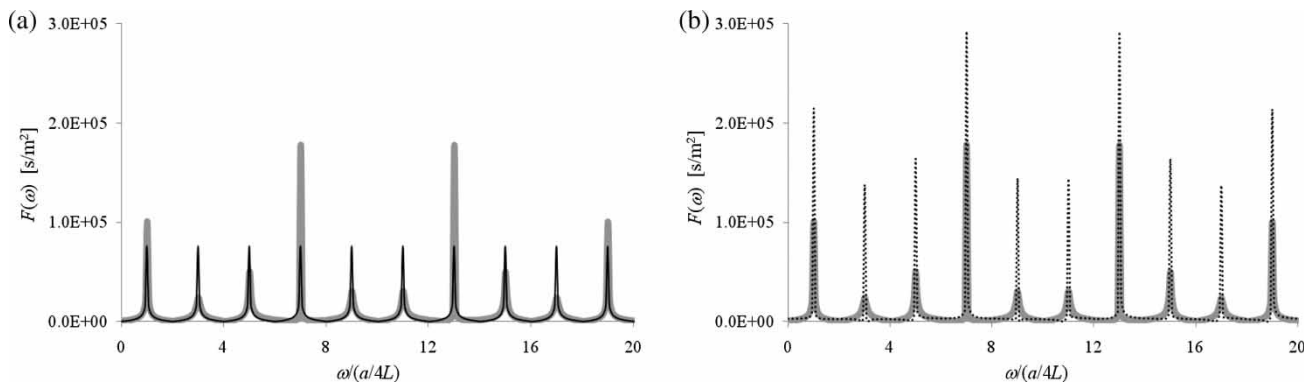


Figure 8 Converted FRF pattern results in the frequency domain by using SRFM. Comparison of full leak information: (a) leak damping only and (b) leak reflection only. —, Leaking pipe response with leak damping and reflection; —, leaking pipe response with leak reflection removed; —, leaking pipe response with leak damping removed

contained in the damping of transients is the leak-induced reflection information. Therefore, for all the transient-based leak detection methods, the transient reflection information behaves relatively much more critical than the damping information.

5 Practical implications

From the analysis, it is concluded that all transient leak detection methods have greater dependence on the presence of leak reflection information. In practical applications, the design of transient signals for fault detections with clear reflections is ideal for this technology. In the past decades, numerous researchers have

worked with different types of transients for condition monitoring, from the usage of pump trips to inline valve closures and to recent solenoid valves (Liou and Tian 1995, Brunone 1999, Souza *et al.* 2000, Tang *et al.* 2000, Lee *et al.* 2006). Those efforts found that the transients generated using the first two methods are generally slow and will create only dispersed reflections that are difficult to identify, whereas the latter is fast and can give clear reflections but is more difficult to set up as additional devices need to be installed to existing pipes. From the present findings, the fast method for generating transients is preferable for transient-based leak detection.

In contrast, the complexities in practical pipe systems such as energy fluxes, junction connections and variations in pipe

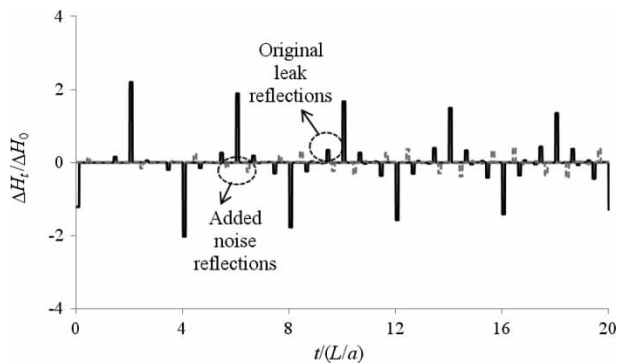


Figure 9 Transient head in the time domain for leaking pipe with additional reflections: —, original complete signal; ---, added noise reflections

geometries through corrosion and sedimentation can affect the transient damping and reflections of the signal traces. The energy fluxes along pipeline increase the signal damping, whereas junction boundaries and changes in pipe geometries generate additional reflections. From the results of the above analysis, the additional reflections (considered as “noise” in the data) may have a great influence on the application of the transient-based leak detection methods. To illustrate this effect, another scenario is conducted by adding additional reflections (to represent “noise” perturbations) to the original signal in Fig. 5. The new contaminated signal is shown in Fig. 9, with the amplitudes of the added reflections comparable to the amplitudes of the real leak reflections.

The results of leak location prediction by the four methods for different leak cases are shown in Table 4. Under this condition, these results indicate that the leak prediction results become inaccurate and the maximum error for each method is over 20% if there is comparable reflection “noise” in the data. Further work is clearly needed to identify fault-induced reflections from other sources of reflection in the system for the practical application of the techniques in reality.

6 Conclusions

This research investigates the impacts of two leak-induced information, damping and reflection, and their relative importance to

the application of present transient-based leak detection methods. Four representative types of leak detection methods, namely, the TRM, the TDM, the system reflection function-based method and the ITM, were applied and discussed.

The results indicate that all transient-based methods are unable to detect and locate the leak in the absence of the leak-reflected signal, whereas the leak is detected and located even in the absence of leak-induced damping. The analysis of this study demonstrates that all techniques hinge on the presence of the leak-reflected signal. Moreover, the information previously thought to be contained in the damping of slow transients are in fact low frequency reflections from the leak and are essential to the leak detection process for all methods. The findings of this study have implications on the design of transient signals for fault detection and suggest that high bandwidth signals with clear reflections are ideal for this technology. Meanwhile, the understanding and definition of system reflection properties can promote the practical applications of transient-based leak detection methods.

Acknowledgements

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Notation

a	=	wave speed
A	=	pipe cross-sectional area
D	=	pipe diameter
f	=	pipe friction factor
$F(\cdot)$	=	frequency response function
g	=	gravitational acceleration
G	=	objective function in Eq. (8)
$\hat{h}$	=	inverted peak head magnitude
H	=	pressure head
H_{tank}	=	pressure head at reservoir/tank
H^m, H^p	=	measured and predicted pressure head
i	=	time-step index
$I(\cdot)$	=	impulse response function
L	=	pipe length

Table 4 Results of considering additional uncertain reflections

Case	Real leak location, $x_L^* r$	TRM		TDM		SRFM		ITM	
		$x_L^* p$	ε (%)	$x_L^* p$	ε (%)	$x_L^* p$	ε (%)	$x_L^* p$	ε (%)
16	0.10	0.23	13.0	0.38	28.0	0.20	10.0	0.29	19.0
17	0.30	0.33	3.0	0.36	6.0	0.35	5.0	0.71	41.0
18	0.60	0.43	−17.0	0.83	23.0	0.33	−27.0	0.51	−9.0
19	0.80	0.53	−27.0	0.59	−21.0	0.61	−19.0	0.35	−45.0
20	0.90	0.63	−27.0	0.67	−23.0	0.46	−44.0	0.57	−33.0

m	=	peak magnitude number
n	=	harmonic mode number
N	=	total number of discrete time steps
Q	=	discharge
Q_L	=	initial leak discharge
Q_V	=	valve discharge
R_{nL}	=	leak-induced damping rate
t, t^*	=	time variables
x	=	spatial coordinate
$x()$	=	input signal function
x_L^*	=	dimensionless leak location
$x_L^*_p$	=	predicted dimensionless leak location
$x_L^*_r$	=	real dimensionless leak location
$X()$	=	Fourier transform of input signal
$y()$	=	output signal function
$Y()$	=	Fourier transform of output signal
ΔH_0	=	initial transient pressure head
ΔH_t	=	immediate transient pressure head
α, β, θ	=	fitting coefficients in Eq. (7)
ε	=	prediction error
ω	=	frequency

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